

6(a)	$(E =) Q / 4\pi\epsilon_0 r^2$	M1
	where ϵ_0 is permittivity (of free space)	A1
6(b)(i)	field does not change direction/field does not become zero	M1
	so (charges have) opposite (sign)	A1
6(b)(ii)	minimum is at the midpoint (between the charges)	M1
	so (magnitudes are the) same	A1
6(c)	force = field strength $\times$ charge and force = mass $\times$ acceleration or acceleration is proportional to field strength	B1
	(from $x = 3.0$ cm) to $x = 5.0$ cm: acceleration decreases	B1
	at $x = 5.0$ cm: acceleration is a minimum	B1
	from $x = 5.0$ cm (to $x = 7.0$ cm): acceleration increases	B1